International Journal of Heat and Mass Transfer 159 (2020) 120115

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/hmt

Prediction of "critical heat flux" for supercritical water and CO_2 flowing upward in vertical heated tubes

Zufen Luo[a], Xiande Fang[a,*], Yang Yang[a], Weiwei Chen[b], Lisheng Zhang[c]

[a] Key Laboratory of Aircraft Environment Control and Life Support, MIIT, Nanjing University of Aeronautics and Astronautics, 29 Yudao St., Nanjing 210016, China
[b] School of Energy and Mechanical Engineering, Nanjing Normal University, No.2 Xuelin Road, Nanjing 210023, China
[c] 650 Research Institute, AVIC Hongdu Aviation Industry Group Ltd., Nanchang 330024, China

ARTICLE INFO

Article history:
Received 17 December 2019
Revised 10 June 2020
Accepted 21 June 2020
Available online 1 July 2020

Keywords:
Critical heat flux
Heat transfer deterioration
Supercritical flow
Water
CO_2
Correlation

ABSTRACT

The prediction method for the occurrence of supercritical critical heat flux (CHF) is investigated both theoretically and statistically. The supercritical CHF is defined as the lowest heat flux that causes the wall temperature (T_w) peak to occur. The new definition is helpful to avoid the confusion caused by different definitions of the heat transfer deterioration (HTD) in supercritical flow. A new theory to explain supercritical CHF mechanisms is proposed, which considers the rupture of flow stability between the liquid-like and vapor-like layers of the supercritical flow in a heated tube as the main reason for supercritical CHF phenomena. From this theory, a general mathematical model for supercritical CHF identification for upward vertical flow is derived. To substantiate the general mathematical model, two databases for supercritical CHF of water and CO_2 flowing upward in vertical circular tubes are compiled from the available literature, based on which the general model is reduced to a practical correlation applicable to both supercritical water and supercritical CO_2 for CHF occurring before the pseudo-critical point. The new correlation differentiates the CHF cases from the non-CHF ones very well, and it has much higher prediction accuracy than any existing counterpart.

1. Introduction

Supercritical fluids such as supercritical water and supercritical CO_2 have been widely applied in many industrial sectors, such as advanced nuclear reactors [1], ultra-supercritical units [2], and solar-thermal power systems [3,4]. The design and optimization of these new systems require a thorough understanding of the unique thermo-hydraulic characteristics of supercritical fluids.

In flow boiling and pool boiling heat transfer, the critical heat flux (CHF) represents the upper limit for the safe operation of a cooling system that relies on boiling heat transfer. The occurrence of CHF may cause a sudden large temperature rise at the heated surface, which potentially results in its physical burnout. For heat transfer with a fluid at supercritical pressure, although the phase change as seen in boiling heat transfer does not exist, the phenomenon of sharp or broad wall temperature spikes at one or more locations along the heated wall of a channel can occur due to strong variations in the fluid's thermophysical properties around the pseudo-critical point. This sharp or broad spike of wall temperature (T_w) is usually called wall temperature peak, which is a main characteristic of heat transfer deterioration (HTD) phenomena and may cause the failure of the heated surface. Hence, the appearance of T_w peak should be avoided in the safety design and operation of a cooling system with supercritical flow. The present study defines the lowest heat flux which causes T_w peak to occur as CHF of supercritical fluids, denoted by q_{cr}, which is correspondent to that in two-phase boiling from the perspective of similarity. This definition is helpful to avoid the confusion on HTD, as discussed in the following section.

The T_w peak phenomenon, or the supercritical CHF phenomenon, was first observed in the 1960s [5,6]. Since then, many efforts have been made on studying it. However, its mechanisms are still not fully understood and a widely-recognized method for predicting its occurrence has yet to be established [7,8].

Many correlations for predicting supercritical CHF have been proposed. Most of these correlations specified q_{cr} for a specific fluid as a power function of mass flux G, namely as $q_{cr} = CG^n$ [9]. There is an overall agreement that the power law is able to fit individual experimental data well. However, for different fluids and channel geometries, the constants C and n are usually different [5,6,9–21].

* Corresponding author.
E-mail address: xd_fang@yahoo.com (X. Fang).

https://doi.org/10.1016/j.ijheatmasstransfer.2020.120115

Nomenclature

A	peripheral heat transfer area, m
a	ratio of mass flux
c_p	specific heat, J/ kg K
D	tube diameter, mm, m
Er	error rate of a judgment
F_D	correction factor for the tube diameter
G	mass flux, kg/m^2 s
h	enthalpy, J/kg, kJ/kg
htc	heat transfer coefficient, W/m^2 K, kW/m^2 K
i	latent heat of supercritical fluids, J/kg
l	length, m
M	molecular weight, g/mol
m	mass, kg
P	pressure, MPa
q	heat flux, kW/m^2, W/m^2
T	temperature, K, °C
t	time, s
x	distance along the flow direction, m
Nu	Nusselt number
Re	Reynolds number
Pr	Prantdl number
Greek symbols	
β	thermal expansion coefficient, 1/K
λ	thermal conductivity, W/m K
δ	thickness of the vapor layer, m
σ	stress, Pa
Subscripts	
b	bulk fluid
CO_2	supercritical CO_2
cr	critical
db	Dittus−Boelter
g	vapor
I	inertia force
nht	normal heat transfer
pc	pseudo-critical point
pred	predicted value
w	inner wall
water	supercritical water
0	reference point

For the exponent n, the majority of studies took the form of $n = 1$ [5,10,12–14,16,17,21]. For example, based on their heat transfer experiments of supercritical CO_2, Zhu et al. [21] proposed a correlation as $q_{cr} = 5.126 \times 10^{-4} G h_{pc}$, where h_{pc} is the enthalpy at the pseudo-critical temperature, T_{pc}. A couple of studies for supercritical CO_2 gave the form of $n = 2$ [9,15]. For example, the Kim et al. [15] model for supercritical CO_2 is of the form $q_{cr} = 0.0002G^2$. This correlation might predict the experimental data for supercritical CO_2 better than other relations but could not be extended to supercritical water at various working conditions [7,22]. Based on their supercritical CO_2 experimental investigations, Kline et al. [9] obtained a model similar to that of Kim et al. [15], which is of the form $q_{cr} = 0.0002F_{D,cr}G^2$, where $F_{D,cr} = [0.7+(1\text{-}0.7)/(1+e^{4/(D/8\text{-}2.35)})]$, which is a correction factor for the tube diameter D. They indicated that their correlation had been validated only for CO_2 flows with diameters from 4.6 to 22 mm, mass fluxes from 300 to 1500 kg/m^2 s, and a reduced pressure of 1.13. There are a few correlations with the exponent n not being an integer [6,11,18,19,20], which are just the data fitting results of experimental data.

Many efforts were made to explore the effects of the related factors, such as buoyancy, flow acceleration, and flow direction on supercritical CHF mechanisms [23]. Many researchers [22,24–28] indicated that at relatively low mass flux G and high heat flux q, the buoyancy effect was a main reason for supercritical CHF phenomena. Palko and Anglart [25] indicated that for the high G, buoyancy influence could be neglected and there should exist other mechanisms. Koshizuka et al. [26] conducted a numerical analysis of heat transfer to supercritical water flowing upward in a vertical tube. The results indicate that the HTD occurred due to two mechanisms depending on mass flux G. When G is small, buoyancy force accelerates the flow velocity near the wall, making the flow velocity distribution flat, described as laminarization by some other researchers [27,28] who supported this explanation, and generation of turbulence energy by shear stress reduced. When G is large, viscosity increases locally near the wall by heating, which makes the viscous sublayer thicker and the Prandtl number smaller, reducing the heat transfer. Although the buoyancy and acceleration influences are accepted by many scholars, large discrepancies exist in the process of quantifying these effects [23]. Pis'menny et al. [29] studied experimentally the heat transfer to supercritical water flowing upward and downward in vertical tubes. It was found that the HTD regime started earlier in the upward flow than that in the downward flow because of concurrent and countercurrent effects of free and forced convections in upward and downward flows respectively and free convection deteriorating heat transfer in upward flow.

Many scholars [30–33] studied supercritical heat transfer using the two layer theory. There are different descriptions about the two layers. One description assumes that a turbulent supercritical fluid flow in a pipe consists of a near-wall boundary layer and a turbulent core layer [30–32]. The boundary layer may be a viscous sublayer [30,31], or a turbulent layer [32]. Also, there is a description that a turbulent flow in a pipe consists of a near-wall viscous sublayer, a buffer layer, and a turbulent core [33], where the two layers are referred to as the viscous sublayer and the buffer layer. Pandey et al [30] improved a two layer model by inclusion of the buoyancy and acceleration effects, and the results showed that their model is improved qualitatively in terms of capability to predict HTD.

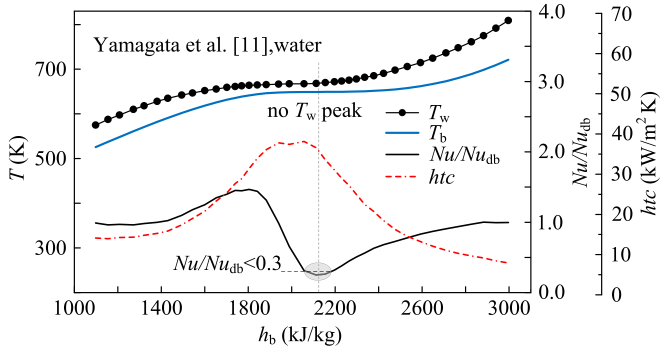
Zhu et al. [21] proposed a two layer theory to describe CHF. They stated that when $T_w > T_{pc}$, before bulk fluid reaches T_{pc}, the tube cross-section contains a vapor-like layer near the wall and a liquid-like layer in the core. Then, they extended the subcritical boiling number to the supercritical pseudo-boiling and proposed a correlation to predict supercritical CHF.

Despite the efforts made by the scientific community, large discrepancies exist in quantifying supercritical CHF, and no supercritical CHF correlation has been found to be widely accepted. In this article, a new hypothesis related to the rupture of flow stability between the liquid-like and vapor-like layers is proposed to explain supercritical CHF phenomena, from which a general form of supercritical CHF model for upward vertical flow is derived. Then, an experimental database is compiled from the available literature for supercritical water and CO_2 flowing in vertical tubes, based on which a unified correlation to predict supercritical CHF is proposed to predict the CHF occurring before the pseudo-critical point for both supercritical water and supercritical CO_2.

2. Discussion on supercritical CHF

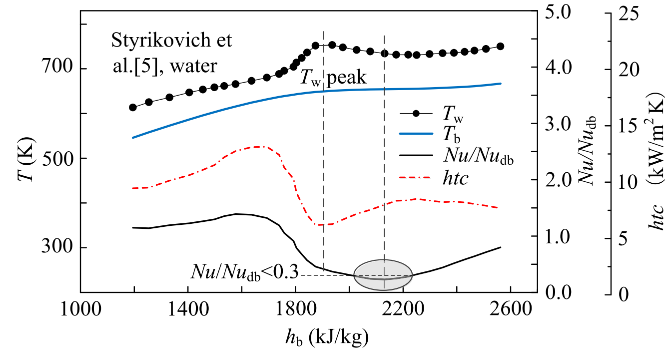
2.1. HTD concept of supercritical flow

The supercritical CHF phenomenon discussed here is sometimes called an HTD phenomenon by others. However, the HTD concept of supercritical flow is not that clear. Pioro et al. [34] reviewed the



P = 22.6 MPa, G = 1180 kg/m^2 s, q = 698 kW/m^2, D = 10 mm.

Fig. 1. HTD case judged by $Nu/Nu_{db} < 0.3$ criterion [35].



P = 24 MPa, G = 700 kg/m^2 s, q = 640 kW/m^2, D = 22 mm.

Fig. 2. Comparison of HTD cases judged by $Nu/Nu_{db} < 0.3$ and T_w peak criteria [35].

literature of HTD of supercritical water. Zhang [35] classified HTD definitions into three types:

Type 1 is based on T_w peak. If the wall temperature has peak(s) along the tube, the case is considered as HTD. If T_w increases monotonously, the case is deemed to be a non-HTD one. Among the three types, this type is the most direct description of HTD phenomena [9,12,15,21,35].

Type 2 is based on heat transfer comparison. There are two commonly-seen criteria in this category. The most frequently used one is if $Nu/Nu_{db} < 0.3$, then the heat transfer is deteriorated, where Nu is the actual Nusselt number, and Nu_{db} is the Nusselt number determined by the Dittus–Boelter [36] Eq. (1). The other is if $htc/htc_0 < 1.0$, then the heat transfer is deteriorated, where htc is the actual heat transfer coefficient (HTC), and htc_0 is the HTC usually calculated using the Dittus–Boelter [36] equation with the thermophysical properties valued at the reference point (Eq. (2)).

$$Nu_{db} = 0.023Re^{0.8}Pr^{0.4} \tag{1}$$

$$htc_0 = 0.023Re_0^{0.8}Pr_0^{0.4}\frac{\lambda_0}{D} \tag{2}$$

where Re is the Reynolds number, Pr is the Prantdl number, λ is the thermal conductivity (W/m K), D is the tube inner diameter in m, and subscript 0 means the thermophysical properties are valued at the reference point, usually taking $h_b = 840$ kJ/kg for water and $h_b = 120$ kJ/kg for CO_2, where h_b is the enthalpy valued at the bulk fluid temperature.

Type 3 is based on the upper limit of T_w. A heat transfer is considered to be deteriorated if T_w exceeds the temperature limit, which is dependent on the material of the heated tubes.

The HTD cases identified by different HTD criteria can be very different, which causes confusion. In Fig. 1, the shadow region is the one with $Nu/Nu_{db} < 0.3$, which might be inappropriate to be called HTD since the correspondent HTC is relatively big. When using the $Nu/Nu_{db} < 0.3$ criterion, the resultant HTD region may locate around the pseudo-critical point. Due to the positive heat transfer characteristics of a supercritical fluid near the pseudo-critical point, the Nu/Nu_{db} may sometimes go down below 0.3, while the actual HTC is still high, even greater than those within the normal heat transfer (NHT) region of the channel. Therefore, the value of Nu/Nu_{db} just reflects the difference between the real HTC and that calculated with the Dittus–Boelter correlation. This method is often used to segment supercritical heat transfer data for developing a heat transfer correlation for $Nu/Nu_{db} < 0.3$ region [37,38]. When calling the $Nu/Nu_{db} < 0.3$ region HTD region, one should keep in mind that the real HTC in that region may not be low.

Fig. 2 compares HTD cases judged by the $Nu/Nu_{db} < 0.3$ and T_w peak criteria, showing inconsistent results. The T_w peak point falls into the weakest heat transfer region, which should be the real HTD, while in the $Nu/Nu_{db} < 0.3$ region, the heat transfer is still fairly good. Considering the $Nu/Nu_{db} < 0.3$ region is also called HTD [14,18,19,38], the present study defines T_w peak phenomena as CHF phenomena.

2.2. Type of supercritical CHF

According to the location of appearance, supercritical CHFs can be classified in three categories as follows:

(a) CHF occurring before the pseudo-critical point. Before the bulk fluid temperature T_b reaches T_{pc} (with $T_b \le 0.995T_{pc}$), one or more T_w peaks occur along the test section (Fig. 3a);
(b) CHF occurring after the pseudo-critical point. When T_b is close to or higher than T_{pc} (with $T_b > 0.995T_{pc}$), the T_w peak starts to be observed (Fig. 3b);
(c) CHF occurring both before and after the pseudo-critical point (Fig. 3c).

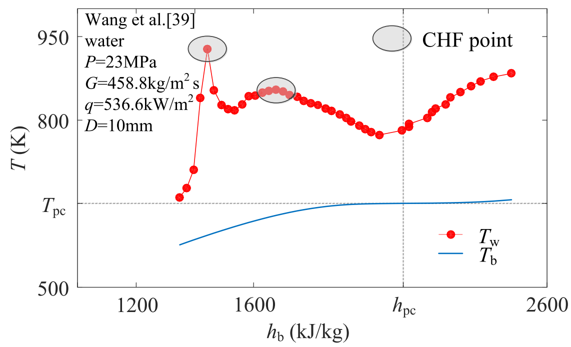
The classification mentioned above is just for a better understanding of CHF phenomena. Considering that when T_b is close to or higher than T_{pc}, the wall temperature T_w will rise quickly in the subsequent heat transfer process due to the deterioration of supercritical fluid heat transfer characteristics, most experiments paid little attention to the HTD phenomena occurring after the pseudo-critical point, and thus the corresponding available CHF data are rare. Therefore, this paper focuses on CHF phenomena occurring before the pseudo-critical point.

3. General form of CHF correlation

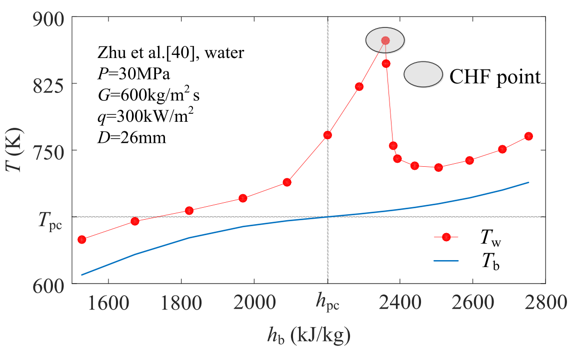
3.1. New theory of CHF mechanisms

In the two layer theory proposed by Zhu et al. [21] to describe supercritical CHF in a heated tube, when $T_w > T_{pc}$, before the bulk fluid reaches T_{pc}, there is a vapor-like layer near the wall and a liquid-like layer in the core. The vapor-like layer has $T > T_{pc}$, the liquid-like layer has $T < T_{pc}$, and the two-layer interface has $T = T_{pc}$. They assumed that the competition between vapor expansion induced momentum force and inertia force dominates the transition from NHT to HTD, and that the vapor expansion effect tends to increase the vapor-like layer thickness.

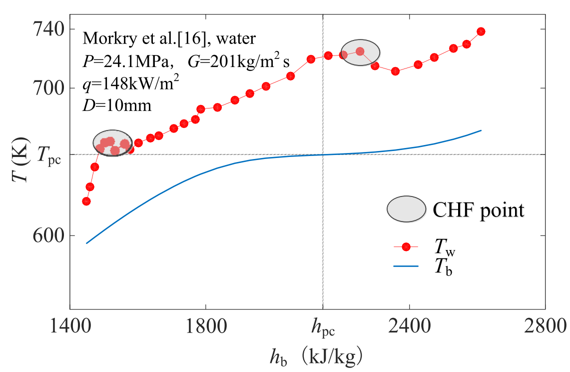
Extending the Zhu et al. [21] two-layer theory, the present paper proposes that the rupture of flow stability between the two layers is the main reason for supercritical CHF phenomena. In the NHT case, the heat and mass transfer between the two layers is mild and stable, which leads to a mild rise of the wall temperature and the vapor-like layer thickness along the heated tube. When approaching CHF, however, the wall heat flux is excessive, which



(a) CHF occurring before pseudo-critical point



(b) CHF occurring after pseudo-critical point



(c) CHF occurring before and after pseudo-critical point

Fig. 3. Types of supercritical CHF phenomena.

leads to a rapid growth in the vapor-like layer thickness. Due to the poor heat transfer characteristics of the vapor-like fluid, the heat transfer from the vapor-like layer to the liquid-like layer gets more difficult, resulting in a rapid rise in wall temperature. When the vapor-like layer thickness grows to a critical value, the flow stability of the two layers is disrupted, causing the two fluids to mix and the vapor-like layer to get thinner, which in turn improves the heat transfer near the wall and leads to a decrease in wall temperature, forming a wall temperature peak.

Below, a general form of supercritical CHF correlation is derived based on the new theory.

3.2. Development of general form of CHF correlation

For a stable supercritical flow, a micro-segment with a dimension along the flow direction of dx is taken into consideration, as illustrated in Fig. 4, where the dashed line is the vapor-liquid interface whose temperature equals the pseudo-critical temperature

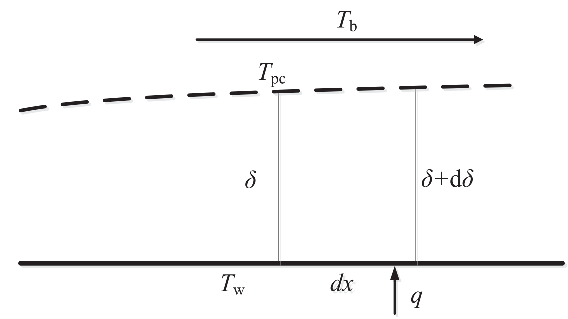


Fig. 4. Flow and heat transfer close to the wall for supercritical fluids.

T_{pc} [21]. The heat flux q acts on the wall, and the wall temperature T_w is above T_{pc}. Close to the wall is the vapor-like layer with a thickness of δ. The temperature of the liquid-like core is approximated to T_b, which is the bulk temperature. Actually, the temperature of the liquid-like core is slightly smaller than T_b. However, as the vapor-like layer is a very small amount compared to the liquid-like core, this approximation is reasonable.

The liquid-like fluid transforms into the vapor-like one through absorbing heat. When considering the transformation between the two fluids, the parameter group $c_{p,pc}/\beta_{pc}$ is used as the latent heat of supercritical fluids at the pseudo-critical point [17], where c_p is the isobaric specific heat and β is the thermal expansion coefficient, the subscript pc means the parameter is valued at the pseudo-critical point. The latent heat of a supercritical fluid, i_{pc}, corresponds to the heat absorption for a unit change of the fluid at pseudo-critical point is of the form

$$i_{pc} = \frac{c_{p,pc}}{\beta_{pc}} \tag{3}$$

The heat taken from the wall transfers across the "vapor-liquid" interface to the liquid-like side. It has the potential to increase the vapor-like layer thickness due to the "phase change" process occurring at the "vapor-liquid" interface. The mass increase rate m'_1 can be derived by assuming a stagnation of the fluid flow

$$m'_1 = \frac{dm_1}{dt} = \frac{q}{i_{pc}} dA \tag{4}$$

where m_1 is the mass increase of the vapor-like fluid generated by the heat flux, A is the peripheral heat transfer area of the micro-segment with the length of dx, and t is the time.

On the other hand, the inertia force σ_I, which is mainly from the liquid-like fluid flow [41], has a taking away effect on the vapor-like layer with turbulent diffusion mass transfer. The mass change rate m'_2 of the vapor-like fluid in the micro-segment due to the inertia force effect, which is suggested to be linearly dependent on the inertia force [42], can be expressed by

$$m'_2 = CG^2 dA \tag{5}$$

where C is the parameter affected by the factors including the inclination of the "vapor-liquid" interface to the inner wall and the turbulence shear stress on the interface.

The difference between m'_1 and m'_2, denoted by $\Delta m'$, is the net mass increase rate of the vapor-like fluid, which causes a growth of the vapor-like layer thickness.

$$\Delta m' = \left(\frac{q}{i_{pc}} - CG^2\right) dA = G_g dA_\delta \tag{6}$$

where G_g is the equivalent mass flux of the vapor-like fluid corresponding to the thickness growth $d\delta$, which is much smaller than G, and A_δ is the projected peripheral area of $d\delta$ normal to the flow direction.

Table 1
Experimental data sources of supercritical water flowing upward in circular heated tubes [38].

Source	Flow range: P(MPa)/G(kg/m^2 s)/q(kW/m^2)	D(mm)	Number of points
Shiralkar and Griffith [6]	22.7,23.4/461-610/252.4-2328.6	10,16	7
Vikhrev et al. [10]	26.5/495,1400/507-930	20.4	3
Yamagata et al. [11]	24.6/1260/233-698	7.5,10	4
Mokry et al.[16,37]	24/201-1500/129-1500	10	18
Wang et al. [39]	23-26/450-1200/200-1200	10	17
Zhu et al. [40]	23-30/600,1200/200,300	26	6
Pan et al. [43]	22.5-30/1009-1626/216-822	17	10
Alekseev et al. [44]	24.5/650/560	10.4	1
Griem [45]	22,25/500-2500/300-600	14	6
Li et al. [46]	23-26/459.8-1497.5/192-1326.5	7.6	23
Li et al. [47]	23-25/655-1263/466-989	6	13
Shitsman [48]	24.5/354-608/290-580	8,16	9
Swenson et al. [49]	22.8,31/2150/789,1735	9.42	3
Xu et al. [50]	23/1200/200-600	12	3
Lei et al. [51]	23-28/600-1200/200,300	26	7
Mörk-Mörkenstein and Herkenrath [52]	24/1500/800-1200	10	5
Gu et al. [53]	23,25/600, 1000/700,1000	7.6,10	6

It is proposed that the vapor-like layer of the micro-segment has a critical thickness, δ_{cr}. When $\delta < \delta_{cr}$, the supercritical fluid flow and heat transfer is in a stable state. When δ reaches δ_{cr}, the stable state starts to collapse, causing a huge mixing of the two fluids near the interface, the vapor-like layer gets thinner in the subsequent heated section, and the wall temperature goes down. As a result, a wall temperature peak is generated at the critical point. Since δ is very thin, the difference between the peripheral length of the interface and that of the wall heat transfer area can be neglected. Therefore, Eq. (6) can be reduced to

$$\left(\frac{q}{i_{pc}} - CG^2\right)dx = G_g d\delta \tag{7}$$

Integrating the above equation at the position of the critical point yields

$$\int_0^{\delta_{cr}} d\delta = \int_0^l \left(\frac{q}{i_{pc}} - CG^2\right)\frac{dx}{G_g} \tag{8}$$

where l is the length from the position at which T_w equals T_{pc} to the position at which δ equals δ_{cr}.

If the variation of C along the heated channel is negligible, then G_g is nearly constant. Considering if $T_w = T_{pc}$, then $\delta = 0$ and $l = 0$, and at q_{cr} condition, if $T_w = T_{peak}$, then $\delta = \delta_{cr}$ and $q=q_{cr}$, the approximate solution of Eq. (8) can be of the form

$$\delta_{cr} G_g = \left(\frac{q_{cr}}{i_{pc}} - CG^2\right)l \tag{9}$$

or

$$q_{cr} = C i_{pc} G^2 + K \tag{10}$$

where

$$K = \delta_{cr} i_{pc} G_g / l \tag{11}$$

Substituting $a_g = G_g\,/G$ into Eq. (11), it follows that

$$K = a_g \delta_{cr} i_{pc} G / l \tag{12}$$

or

$$\frac{K}{q_{cr}} = a_g \delta_{cr} i_{pc} \frac{G}{q_{cr}} \frac{1}{l} \tag{13}$$

The orders of magnitude of a_g, δ_{cr}, i_{pc}, and l need to be analyzed to obtain the order of magnitude of K/q_{cr}, and thus K.

The length of the heated tube usually has an order of magnitude of 1 m, and the order of magnitude of l can be estimated at 10^{-1} m. Through the statistics of the experimental data collected in the present paper (Table 1), the i_{pc} has an order of magnitude of 10^5 J/kg, and G/q_{cr} has an order of magnitude of 10^{-3}–10^{-2}kg/J.

Knowing that q_{cr} is defined as the lowest heat flux that causes the wall temperature (T_w) peak to occur, there should exist at least one heat flux that is greater than q_{cr}, denoted by $q_{cr}{}^+$. At $q_{cr}{}^+$, the maximum vapor-like layer thickness, $\delta_{cr}{}^+$, is no less than δ_{cr}. As the vapor-like layer is a very thin boundary close to the wall, its bulk velocity is small, and thus the conducting sublayer assumption used by Pandey et al. [30,31] is adopted here. With this assumption, it follows that

$$q_{cr}^+ = \frac{k_g (T_{w,peak}^+ - T_{pc})}{\delta_{cr}^+} \tag{14}$$

where k_g is the thermal conductivity of the vapor-like layer evaluated at temperature $T_g = 0.5\ (T_{w,peak}^+ + T_{pc})$, and $T_{w,peak}^+$is the peak wall temperature corresponding to $q_{cr}{}^+$. Based on Eq. (14) and the experimental data collected in the present paper, the $\delta_{cr}{}^+$ has an order of magnitude of 10^{-5} m so that the δ_{cr} has an order of magnitude no bigger than 10^{-5} m.

Since the vapor-like layer is a very thin boundary close to the wall, its velocity should be less than the half of the bulk velocity, thus $Gg < 0.5G(\rho_g/\rho_b)$, where $0.5 \times \rho_g/\rho_b$ has an order of magnitude of 10^{-1}. Therefore, a_g has an order of magnitude no bigger than 10^{-1}.

To sum up, under the referred experimental conditions in the present paper, it can be estimated that

$$K/q_{cr} \le 10^{-1} \times 10^{-5} \times 10^5 \times \left(10^{-3} - 10^{-2}\right)/10^{-1} = 10^{-3} - 10^{-2}$$

That is to say, K is (10^{-3}–10^{-2}) times less than the first term of the right hand of Eq. (10).

Due to lacking the sufficient knowledge about C and K,Eq. (10) is just a result of theoretical analysis and cannot be applied to engineering design directly, and the rationality of simplifying C needs to be tested. A practical way is to use a statistical method to finalize C and K based on the available experimental data. For this reason, the following section presents the experimental CHF databases of supercritical water and CO_2 compiled from the available literature, after which the database is used to substantiate Eq. (10) to propose a practical and uniform CHF correlation for supercritical water and supercritical CO_2.

4. Experimental databases for supercritical CHF of water and CO_2

The CHF experimental data sources for supercritical water were taken from literature [38], as listed in Table 1. All the experimental data were obtained from supercritical water flowing upward in circular heated tubes. The parameter ranges of the data sources cover mass flux from 201 to 2500 kg/m^2 s, heat flux from 129 to 2328.6 kW/m^2, pressure from 22 to 31 MPa, and tube hydraulic diameter from 6 to 26 mm.

The CHF experimental data sources for supercritical CO_2 are listed in Table 2, and all the experimental data were obtained from supercritical CO_2 flowing upward in circular heated tubes. The parameter ranges of the data sources cover mass flux from 100 to 2000 kg/m^2 s, heat flux from 2.9 to 436 kW/m^2, pressure from 7.5 to 21 MPa, and tube hydraulic diameter from 2 to 22 mm.

5. New correlation for CHF of supercritical flow

5.1. Development of new correlation

The genetic algorithm is used to find the two unknown parameters C and K in Eq. (10). The predicted q_{cr}, denoted by $q_{cr,pred}$, can be calculated with a combination of C and K. For a given case, whose actual heat flux is q, the prediction is judged to be right or wrong based on the following rule:

(a) If $q_{cr,pred} \leq q$ and the given case is a CHF case, or if $q_{cr,pred} > q$ and the given case is a non-CHF case, then the prediction is judged to be right, and the sample is denoted by $f(C,K) = 0$;
(b) If $q_{cr,pred} \leq q$ and the given case is a non-CHF case, or if $q_{cr,pred} > q$ and the given case is a CHF case, then the prediction is judged to be wrong, and the sample is denoted by $f(C,K) = 1$.

Define the error rate as

$$Er = \frac{1}{N}\sum_{N} f(C,K) \tag{15}$$

with

$$f(C,K) = \begin{cases} 0 \text{Rightprediction} \\ 1 \text{Wrongprediction} \end{cases} \tag{16}$$

where N is the number of the samples (experimental cases). The optimization objective is then defined as

$$MinEr = \min\left(\frac{1}{N}\sum_{N} f(C,K)\right) \tag{17}$$

For a given number of experimental cases, different combinations of C and K correspond to different Er's. With the global random hunting function of the genetic algorithm, a proper combination of C and K satisfying Eq. (17) exists.

At first, the proper combination of C and K was trained by the experimental data. The result showed that $K = 0.0063$ kW/m^2, while $Ci_{pc}G^2 > 10$ kW/m^2, indicating that the order of magnitude analysis result that K is (10^{-3}–10^{-2}) times smaller than $Ci_{pc}G^2$ is conservative and thus reliable. Since K is negligible, in the second round of optimization process, only C is considered, which is trained by both the supercritical water and CO_2 experimental data. The final result is

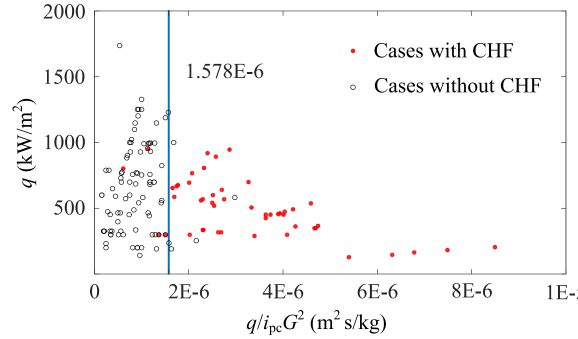


Fig. 5. CHF occurrence prediction for supercritical water.

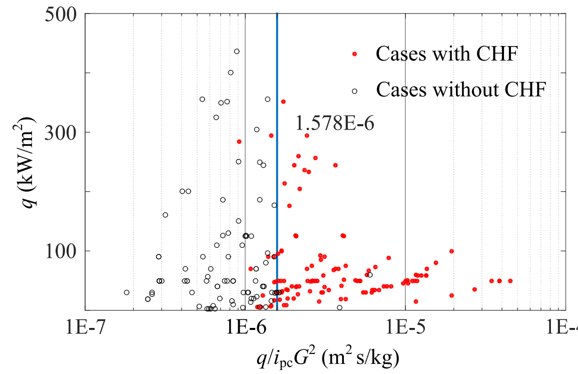


Fig. 6. CHF occurrence prediction for supercritical CO_2.

$$q_{cr} = 1.578 \times 10^{-6} i_{pc} G^2 \tag{18}$$

Figs. 5 and 6 show the new correlation predictions about the CHF occurrences of supercritical water and CO_2, respectively. As expected, the result agrees with supercritical CO_2 very well, as can be seen from Fig. 6. Therefore, Eq. (18) can predict well the CHF cases occurring before the pseudo-critical point. Strictly speaking, however, C in Eq. (18) is unlikely to be a constant, and more efforts are still needed to study it.

5.2. Comparison of the new correlation with existing correlations

The new correlation is compared with the existing correlations based on the database shown in Tables 1 and 2, with Er defined by Eq. (15) as the criterion. The results are listed in Table 3, where M is the molecular weight and the subscripts nht, water and CO_2 denote normal heat transfer, supercritical water, and supercritical CO_2, respectively.

Table 2
Experimental data sources of supercritical CO_2 flowing upward in circular heated tubes.

References	Flow range: P(MPa)/G(kg/m^2 s)/q(kW/m^2)	D(mm)	Number of points
Shiralkar and Griffith [6]	7.76/406.8,915/15.7,56.15	10.2	2
Kline et al. [9]	8.32/300-1000/18-260	4.6,8,22	8
Zhu et al. [21]	7.8-21/519-1600/92-355	10	19
Gupta et al. [54]	8.36-8.8/784-2000/18.4-235.9	8	7
Eter et al. [55]	7.6-8.36/192-1184/2.9-126.3	8	54
Zahlan et al. [56]	7.52-8.58/200-2000/30-436	8, 22	33
Kim et al. [57]	7.75-8.85/400-1200/20-150	4.4	30
Kim et al. [58]	8.12/400,1200/30,50	4.4, 9	9
Zhang et al. [59]	7.5/300,400/20-75	16	8
Liu et al. [60]	7.61/901.8/175.9-256.2	10	4
Lei et al. [61]	7.5-10.5/100,200/5-100	5	14
Bae et al. [62]	7.75,8.12/285-1200/30-70	6.32	19
Li et al. [63]	8.8/315/7-52	2	7

Table 3
Existing CHF identification correlations for supercritical fluids upward-flowing in heated tubes.

Authors	Criterion of HTD	Fluids	Identification method of HTD used in study	Er_{water} (%)	Er_{CO2} (%)
Kline et al. [9]	$q_{cr} = 0.0002G^2\ [0.7+0.3/(1+e^{4/(D/8-2.35)})]$	CO_2	$Nu < Nu_{nht,}$ cooperating with the observation of the appearance of T_w peak	57.4	10.7
Styrikovich et al. [5]	$q_{cr} = 0.5815G$	Water	$T_w <$ about 580 °C	31.2	55.0
Shiralkar and Griffith [6]	$q_{cr} = 0.00316G^{1.495}$	CO_2	$Nu/Nu_{db} < 0.5$	65.2	22.4
Yamagata et al. [11]	$q_{cr} = 0.2G^{1.2}$	Water	Not provided	13.5	57.0
Grabezhnaya and Kirillov [13]	$q_{cr} = 0.6G(M_{water}\ /M)$	Multi-fluid	Not provided	29.1	44.4
Cheng et al. [14]	$q_{cr} = 0.001354G\ i_{pc}$	Water	$Nu/Nu_{db} < 0.3$	21.3	32.2
Kim et al. [15]	$q_{cr} = 0.0002G^2$	CO_2	Observation of the appearance of T_w peak	56.7	11.2
Mokry et al. [16]	$q_{cr} = -58.97+0.745G$	Water	$htc/htc_0<1.0$	23.4	53.7
Urbano and Nasuti [17]	$q_{cr} = (1.8 \times 10^{-3}i_{pc}\text{-}91.9)G$	CH4	$(dT_w/dx)_{\ min} < 0$	23.4	22.9
Li et al. [18]	$q_{cr} = D(0.36G/D\text{-}1.1)^{1.21}$	Water	$Nu/Nu_{db} < 0.3$	14.9	55.1
Schatte et al. [19]	$q_{cr} = 1.942 \times 10^{-6}G^{0.795} \times (30\text{-}D)^{0.339}{i_{pc}}^{2.065}$	Water	$Nu/Nu_{db} < 0.3$	29.8	39.3
Zhu et al. [21]	$q_{cr} = 5.126 \times 10^{-4}Gh_{pc}$	CO_2	$Nu < Nu_{nht,}$ cooperating with the temperature overshoot >8°C	30.5	36.0
The new correlation	$q_{cr} = 1.578 \times 10^{-6}i_{pc}G^2$	CO_2 and water	Observation of the appearance of T_w peak	6.4	7.9

Table 3 shows that the new correlation is much better than the existing ones, and that it is well applicable for both supercritical water and supercritical CO_2. From Table 3, it can be seen that most existing correlations do not perform well. The main reasons are as follows:

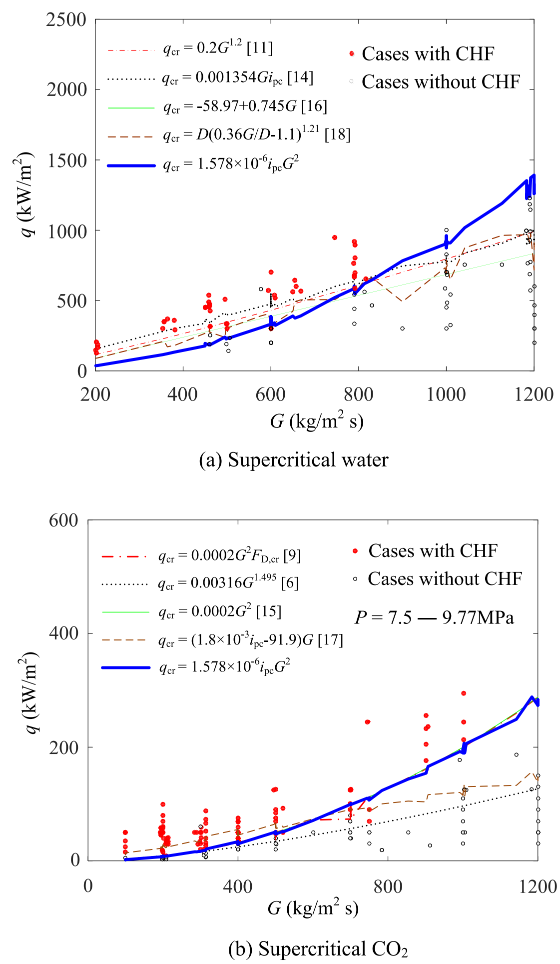
(1) Except the Kline et al. [9] and Kim et al. [15] correlations, all correlations listed in Table 3 were based on other criteria than the T_w peak criterion, and those criteria are not equivalent to the latter, as discussed in Section 2;
(2) Except the Grabezhnaya and Kirillov [13] correlation, all correlations listed in Table 3 were based on just a single fluid, lacking the capability to predict other fluids;
(3) The data some correlations based are limited with narrow mass flux range, leading to the linear or close to linear approximation to be sufficient.

Our purpose to assess the existing correlations is to compare the new correlation with the existing ones, which cannot be construed to rank those existing correlations.

Fig. 7 presents a comparison of the top 4 existing correlations in prediction accuracy with the new one. The data sources were listed in Tables 1 and 2. Due to the impacts of other parameters such as pressure and diameter in the correlations, the curves of q_{cr} versus G for most correlations in Fig. 7 are not that smooth.

Fig. 7 also shows that when the mass flux G varies in a relatively narrow range, for example 200~800 kg/m^2 s for supercritical water in Fig. 7a, it seems not important for the exponent of mass flux to be chosen as 1 or 2, or any other value between 1 and 2. As long as the other parameters match well, a correlation with relatively high prediction accuracy can still be obtained through data fitting. However, when the mass flux G varies in a broad range, the exponent of mass flux would be better to be 2 for both supercritical water (Fig. 7a) and supercritical CO_2 (Fig. 7b). This tendency is more obvious for supercritical CO_2, and the Kim et al. [15] correlation that has the exponent of 2 has received good feedback in previous studies [7,22].

Because there are no extra experimental data available, an attempt to comparison of the new correlation with the top 4 existing CO_2-specific correlations in prediction accuracy, as listed in Table 3 and shown in Fig. 7, is made against 36 data points from [64], which are generated with the direct numerical simulation (DNS). The parameter range covers mass flux from 33 to 233 kg/m^2 s, heat flux from 5 to 30 kW/m^2, pressure from 8 to 8.8 MPa, and tube diameter from 2 to 10 mm. The comparison results show that the present correlation still maintains the minimum Er of 8.3%, and the Er values for the top 4 CO_2-specific correlations of Kim et al. [15], Shiralkar and Griffith [6], Kline et al. [9], and Urbano and Nasuti [17] are 8.3%, 8.3%, 11.3%, and 30.6%, respectively.



(a) Supercritical water

(b) Supercritical CO_2

Fig. 7. Comparison of different correlations for CHF cases.

6. Conclusions

Supercritical CHF phenomena are investigated through theoretical analysis and data fitting methods. The supercritical CHF is defined as the lowest heat flux that causes T_w peak to occur. Traditionally, a T_w peak phenomenon was often called a heat transfer deterioration (HTD). Defining a T_w peak as the occurrence of a supercritical CHF can separate it from the other HTD definitions, such as $Nu/Nu_{db} < 0.3$ and $htc/htc_0 < 1.0$, which is helpful to avoid the confusion caused by different HTD definitions.

The brief discussion and analysis of the HTD phenomena judged by the criteria of T_w peak and $Nu/Nu_{db} < 0.3$ are presented. Using $Nu/Nu_{db} < 0.3$ as a HTD criterion may result in a false conclusion, because the Nu/Nu_{db} may sometimes go down below 0.3 while the actual HTC is still high, even greater than those within the NHT region due to the positive heat transfer characteristics of a supercritical fluid near the pseudo-critical point.

A new theory of CHF mechanisms was proposed, which deems that for supercritical flow in heated tubes, the flow and heat transfer instability of two different layers, i.e. liquid-like layer and vapor-like layer, is the main reason for supercritical CHF phenomena. Based on this theory, a general mathematical model for supercritical CHF identification was derived.

Two databases for supercritical CHF of water and CO_2 flowing upward in vertical circular tubes were compiled from the available literature, based on which the general supercritical CHF identification model is reduced to a practical correlation as $q_{cr} = 1.578 \times 10^{-6} i_{pc} G^2$, which is applicable to both water and CO_2. Since it is based on the experimental data from upward vertical flow, it may not be applicable for other flow directions.

The new correlation distinguishes very well between cases with CHF and those without CHF occurring before the pseudo-critical points, and it has much higher accuracy than any existing correlation proposed for identifying CHF phenomena.

It is possible for the new correlation to apply to other fluids flowing in heated tubes under supercritical pressure. However, the applicability of the new correlation to other supercritical fluids needs to be verified when enough experimental data are available.

The exponent n in the power law $q_{cr} = CG^n$ depends on the variation range of mass flux G. If the variation range is narrow, it seems not important for the exponent n to be chosen between 1 and 2. When the mass flux G varies in a broad range, however, it is better to set the value 2 to the exponent n.

Declaration of Competing Interest

None declared.

Acknowledgments

This study is supported by National Natural Science Foundation of China (51576099, 51176074) and funded by the Priority Academic Program Development of Jiangsu Higher Education Institutions.

References

[1] Y. Ahn, S.J. Bae, M. Kim, S.K. Cho, S. Baik, J.I. Lee, J.E. Cha, Review of supercritical CO_2 power cycle technology and current status of research and development, Nucl. Eng. Technol. 47 (2015) 647–661.
[2] X. Ge, Z. Zhang, H. Fan, J. Zhang, D. Bi, Unsteady-state heat transfer characteristics of spiral water wall tube in advanced-ultra-supercritical boilers from experiments and distributed parameter model, Energy, 10.1016/j.energy.2019.116158.
[3] P.X. Jiang, F.Z. Zhang, R.N. Xu, Thermodynamic analysis of a solar-enhanced geothermal hybrid power plant using CO_2 as working fluid, Appl. Therm. Eng. 116 (2017) 463–472.
[4] M.T. Luu, D. Milani, R. McNaughton, A. Abbas, Analysis for flexible operation of supercritical CO_2 Brayton cycle integrated with solar thermal systems, Energy 124 (2017) 752–771.
[5] M. Styrikovich, T.K. Margulova, Z. Miropol'Skii, Problems in the development of designs of supercritical boilers, Therm. Eng. 14 (6) (1967) 5–9.
[6] B.S. Shiralkar, P. Griffith, The deterioration in heat transfer to fluids at supercritical pressure and high heat fluxes, DSR-70332-55, Cambridge, MIT Engineering Projects Laboratory, Mass, 1968.
[7] D. Huang, Z. Wu, B. Sunden, W. Li, A brief review on convection heat transfer of fluids at supercritical pressures in tubes and the recent progress, Appl. Energy 162 (2016) 494–505.
[8] M. Pizzarelli, The status of the research on the heat transfer deterioration in supercritical fluids: A review, Int. Commun. Heat Mass Transf. 95 (2018) 132–138.
[9] N. Kline, F. Feuerstein, S. Tavoularis, Onset of heat transfer deterioration in vertical pipe flows of CO_2 at supercritical pressures, Int. J. Heat Mass Transf. 118 (2018) 1056–1068.
[10] Y. Vikhrev, Y. Barulin, A. Konkov, A study of heat transfer in vertical tubes at supercritical pressures, Therm. Eng. 14 (9) (1967) 116–119.
[11] K. Yamagata, K. Nishikawa, S. Hasegawa, T. Fujii, S. Yoshida, Forced convective heat transfer to supercritical water flowing in tubes, Int. J. Heat Mass Transf. 15 (12) (1972) 2575–2593.
[12] L. Glushchenko, S. Kalachev, O. Gandzyuk, Determining the conditions of existence of deteriorated heat transfer at supercritical pressures of the medium, Therm. Eng. 19 (2) (1972) 69–72.
[13] V.A. Grabezhnaya, P.L. Kirillov, Heat transfer under supercritical pressures and heat transfer deterioration boundaries, Therm. Eng. 53 (4) (2006) 296–301.
[14] X. Cheng, Y.H. Yang, S.F. Huang, A simplified method for heat transfer prediction of supercritical fluids in circular tubes, Ann. Nucl. Energy 36 (8) (2009) 1120–1128.
[15] J.K. Kim, H.K. Jeon, J.Y. Yoo, J.S. Lee, Experimental study on heat transfer characteristics of turbulent supercritical flow in vertical circular/non-circular tubes, in: Proceedings of the 11th International Topical Meeting on Nuclear Reactor Thermal-Hydraulics (NURETH-11) Avignon, France, October 2005 2–6.
[16] S. Mokry, I. Pioro, A. Farah, K. King, S. Gupta, W. Peiman, P. Kirillov, Development of supercritical water heat-transfer correlation for vertical bare tubes, Nucl. Eng. Des. 241 (4) (2011) 1126–1136.
[17] A. Urbano, F. Nasuti, On the onset of heat transfer deterioration in supercritical coolant flow channels, J. Thermophys. Heat Transf. 27 (2) (2013) 298–308.
[18] Z. Li, D. Zhang, Y. Wu, J. Lu, Q. Liu, A new criterion for predicting deterioration of heat transfer to supercritical water in smooth tubes, in: Proceedings of the CSEE, 2014, pp. 6304–6310.
[19] G.A. Schatte, A. Kohlhepp, C. Wieland, H. Spliethoff, Development of a new empirical correlation for the prediction of the onset of the deterioration of heat transfer to supercritical water in vertical tubes, Int. J. Heat Mass Transf. 102 (2016) 133–141.
[20] D.L. Ma, T. Zhou, B. Li, M.A. Shahzad, Y.P. Huang, An improved correlation on the onset of heat transfer deterioration in supercritical water, Nucl. Eng. Des. 326 (2018) 290–300.
[21] B. Zhu, J. Xu, X. Wu, J. Xie, M. Li, Supercritical “boiling” number, a new parameter to distinguish two regimes of carbon dioxide heat transfer in tubes, Int. J. Therm. Sci. 136 (2018) 254–266.
[22] R.N. Xu, F. Luo, P.X. Jiang, Buoyancy effects on turbulent heat transfer of supercritical CO_2 in a vertical mini-tube based on continuous wall temperature measurements, Int. J. Heat Mass Transf. 110 (2017) 576–586.
[23] J. Xie, D. Liu, H. Yan, G. Xie, S.K.S. Boetcher, A review of heat transfer deterioration of supercritical carbon dioxide flowing in vertical tubes: Heat transfer behaviors, identification methods, critical heat fluxes, and heat transfer correlations, Int. J. Heat Mass Transf. 149 (2020) 119233.
[24] D.E. Kim, M.H. Kim, Experimental investigation of heat transfer in vertical upward and downward supercritical CO_2 flow in a circular tube, Int. J. Heat Fluid Flow 32 (2011) 176–191.
[25] D. Palko, H. Anglart, Theoretical and numerical study of heat transfer deterioration in high performance light water reactor, Sci. Technol. Nucl. Install. 2008 (2008) 405072.
[26] S. Koshizuka, N. Takano, Y. Oka, Numerical analysis of deterioration phenomena in heat transfer to supercritical water, Int. J. Heat Mass Transf. 38 (1995) 3077–3084.
[27] W.B. Hall, J.D. Jackson, Laminarization of a pipe flow by buoyancy forces. ASME paper 69-HT-55, 1969.
[28] M. Sharabi, W. Ambrosini, Discussion of heat transfer phenomena in fluids at supercritical pressure with the aid of CFD models, Ann. Nucl. Energy 36 (2009) 60–71.
[29] E.N. Pis'menny, V.G. Razumovskiy, A.E. Maevskiy, I.L. Pioro, Heat transfer to supercritical water in gaseous state or affected by mixed convection in vertical tubes, in: Proceedings of the ICONE14 Conference, Miami, USA, July 2006 17–20.
[30] S. Pandey, E. Laurien, X. Chu, A modified convective heat transfer model for heated pipe flow of supercritical carbon dioxide, Int. J. Therm. Sci. 117 (2017) 227–238.
[31] S. Pandey, E. Laurien, Heat transfer analysis at supercritical pressure using two layer theory, J. Supercrit. Fluids 109 (2016) 80–86.
[32] J.D. Jackson, Models of heat transfer to fluids at supercritical pressure with influences of buoyancy and acceleration, Appl. Therm. Eng. 124 (2017) 1481–1491.
[33] D.E. Kim, M.H. Kim, Two layer heat transfer model for supercritical fluid flow in a vertical tube, J. Supercrit. Fluids 58 (2011) 15–25.
[34] I.L. Pioro, R.B. Duffey, Experimental heat transfer in supercritical water flowing inside channels (survey), Nucl. Eng. Des. 235 (2005) 2407–2430.

[35] Q. Zhang, H. Li, X. Lei, J. Zhang, X. Kong, Study on identification method of heat transfer deterioration of supercritical fluids in vertically heated tubes, Int. J. Heat Mass Transf. 127 (2018) 674–686.
[36] F.W. Dittus, L.M.K. Boelter, in: Heat Transfer in Automobile Radiators of the Tubular Type, 2, University of California Publications in Engineering, Berkeley, 1930, pp. 443–461.
[37] S. Mokry, I. Pioro, P. Kirillov, Y. Gospodinov, Supercritical-water heat transfer in a vertical bare tube, Nucl. Eng. Des. 240 (2010) 568–576.
[38] W.W. Chen, X.D. Fang, A new heat transfer correlation for supercritical water flowing in vertical tubes, Int. J. Heat Mass Transf. 78 (2014) 156–160.
[39] F. Wang, J. Yang, H.Y. Gu, M. Zhao, H.B. Li, D.H. Lu, Experimental investigation on heat transfer performance of supercritical water in vertical tube, Atom. Energy Sci. Technol. 47 (2013) 933–939 (In Chinese).
[40] X. Zhu, Q. Bi, D. Yang, T. Chen, An investigation on heat transfer characteristics of different pressure steam-water in vertical upward tube, Nucl. Eng. Des. 239 (2009) 381–388.
[41] S.G. Kandlikar, Heat transfer mechanisms during flow boiling in micro channels, J. Heat Transf.-Trans. ASME 126 (2004) 8–16.
[42] A.J. Majda, P.R. Kramer, Simplified models for turbulent diffusion: Theory, numerical modelling, and physical phenomena, Phys. Rep. 314 (1999) 237–574.
[43] J. Pan, D. Yang, Z.C. Dong, T. Zhu, Q.C. Bi, Experimental investigation on heat transfer characteristics of water in vertical upward tube under supercritical pressure, Nucl. Power Eng. 32 (2011) 75–79 (In Chinese).
[44] G.V. Alekseev, V.A. Silin, A.M. Smirnov, V.I. Subbotin, Study of thermal conditions on wall of a pipe during removal of heat by water at a supercritical pressure, High Temp 14 (4) (1976) 683–687.
[45] H. Griem, A new procedure for the prediction of forced convection heat transfer at near-and supercritical pressure, Heat Mass Transf. 31 (1996) 301–305.
[46] H.B. Li, J. Yang, H.Y. Gu, M. Zhao, D.H. Lu, Experimental study of supercritical water flow and heat transfer in vertical tube, Prog. Rep. Chin. Nucl. Sci. Technol. 2 (2011) 874–880 (In Chinese).
[47] Y.L. Li, X.K. Zeng, Z.G. Huang, X. Yan, Y.P. Yan, Z.J. Xiao, Experimental study on heat transfer of supercritical water in simple channels, Nucl. Power Eng. 34 (2013) 101–107 (In Chinese).
[48] M.E. Shitsman, Temperature conditions in tubes at supercritical pressures, Therm. Eng. 15 (1968) 72–77.
[49] H.S. Swenson, J.R. Carver, C.R. Kakarala, Heat transfer to supercritical water in smooth-bore tubes, J. Heat Transf.-Trans. ASME 87 (4) (1965) 477–484.
[50] F. Xu, L.J. Guo, Y.F. Mao, X.E. Jiang, Experimental investigation to the heat transfer characteristics of water in vertical pipes under supercritical pressure, J. Xi'an Jiaotong Univ. 39 (2005) 468–471 (In Chinese).
[51] X.L. Lei, H.X. Li, W.Q. Zhang, N.T. Dinh, Y.M. Guo, S.Q. Yu, Experimental study on the difference of heat transfer characteristics between vertical and horizontal flows of supercritical pressure water, Appl. Therm. Eng. 113 (2017) 609–620.
[52] P. Mörk-Mörkenstein, H. Herkenrath, Wärmeübergang an wasser bei erzwungener strömung im gebiet des kritischen druckes und unter hohen heizflächenbelastungen, Chemie-Ingenier-Technik 1 (5) (1967) 250–253.
[53] H.Y. Gu, M. Zhao, X. Cheng, Experimental studies on heat transfer to supercritical water in circular tubes at high heat fluxes, Exp. Therm. Fluid Sci. 65 (2015) 22–32.
[54] S. Gupta, E. Saltanov, S.J. Mokry, I. Pioro, L. Trevani, D.M. Gillivray, Developing empirical heat-transfer correlations for supercritical CO_2 flowing in vertical bare tubes, Nucl. Eng. Des. 261 (2013) 116–131.
[55] A. Eter, D. Groeneveld, S. Tavoularis, Convective heat transfer in supercritical flows of CO_2 in tubes with and without flow obstacles, Nucl. Eng. Des. 313 (2017) 162–176.
[56] H. Zahlan, D. Groeneveld, S. Tavoularis, Measurements of convective heat transfer to vertical upward flows of CO_2 in circular tubes at near-critical and supercritical pressures, Nucl. Eng. Des. 289 (2015) 92–107.
[57] H.Y. Kim, H. Kim, J.H. Song, B.H. Cho, Y.Y. Bae, Heat transfer test in a vertical tube using CO_2 at supercritical pressures, J. Nucl. Sci. Technol. 44 (3) (2007) 285–293.
[58] H. Kim, H.Y. Kim, J.H. Song, Y.Y. Bae, Heat transfer to supercritical pressure carbon dioxide flowing upward through tubes and a narrow annulus passage, Prog. Nucl. Energy 50 (2008) 518–525.
[59] Q. Zhang, H. Li, X. Kong, J. Liu, X. Lei, Special heat transfer characteristics of supercritical CO_2 flowing in a vertically-upward tube with low mass flux, Int. J. Heat Mass Transf. 122 (2018) 469–482.
[60] S. Liu, Y. Huang, G. Liu, J. Wang, L.K.H. Leung, Improvement of buoyancy and acceleration parameters for forced and mixed convective heat transfer to supercritical fluids flowing in vertical tubes, Int. J. Heat Mass Transf. 106 (2017) 1144–1156.
[61] X. Lei, J. Zhang, L. Gou, Q. Zhang, H. Li, Experimental study on convection heat transfer of supercritical CO_2 in small upward channels, Energy 176 (2019) 119–130.
[62] Y.Y. Bae, H.Y. Kim, D.J. Kang, Forced and mixed convection heat transfer to supercritical CO_2 vertically flowing in a uniformly-heated circular tube, Exp. Therm. Fluid Sci. 34 (8) (2010) 1295–1308.
[63] Z.H. Li, P.X. Jiang, C.R. Zhao, Y. Zhang, Experimental investigation of convection heat transfer of CO_2 at supercritical pressures in a vertical circular tube, Exp. Therm. Fluid Sci. 34 (2010) 1162–1171.
[64] https://www.ike.uni-stuttgart.de/forschung/Ueberkritisches-CO2/dns/ [Accessed on 28 May 2020].